Exercise 6

March 15, 2026

1 Exercise 6

[ ]:
```
import matplotlib.pyplot as plt
import numpy as np
```

[4]:
```
import numpy as np
from urllib.request import urlopen
url = "http://archive.ics.uci.edu/ml/machine-learning-databases/00267/
 ↪data_banknote_authentication.txt"
raw_data = urlopen(url)
dataset = np.loadtxt(raw_data, delimiter=",")
print(dataset.shape)
```

```
(1372, 5)
```

[6]:
```
X=dataset[:,[1,3]]
Y=dataset[:,4]
print(X.shape)
print(Y.shape)
```

```
(1372, 2)
(1372,)
```

1.1 6.1

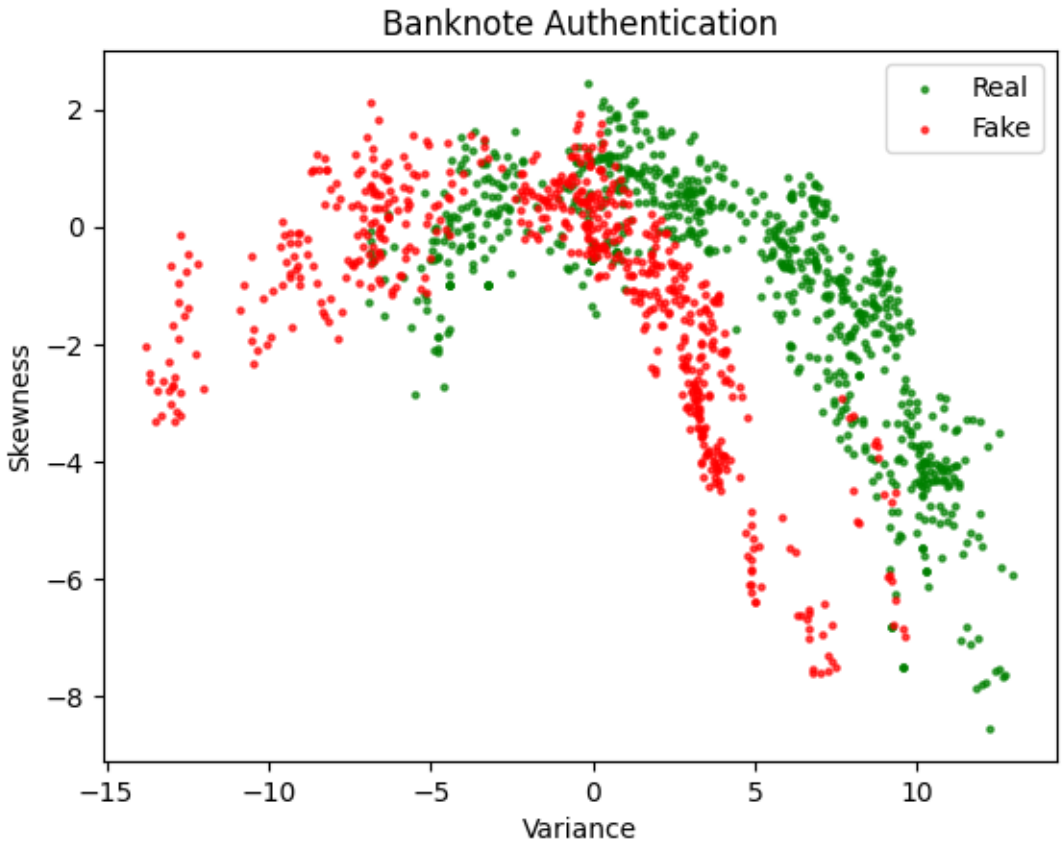
[17]:
```
real = X[Y==0, :]
fake = X[Y==1, :]
plt.scatter(real[:, 0], real[:, 1], color="green", alpha=0.7, label="Real", s=5)
plt.scatter(fake[:, 0], fake[:, 1], color="red", alpha=0.7, label="Fake", s=5)
plt.title("Banknote Authentication")
plt.xlabel("Variance")
plt.ylabel("Skewness")
plt.legend()
plt.show()
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
```



No the real and fake data points cannot be separated with a straight line.

1.2 6.2

[29]:
```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.optimizers import SGD

model = Sequential([
    Dense(1, input_shape=(X.shape[1],), activation="sigmoid")
])

model.compile(optimizer=SGD(learning_rate=0.15),
              loss="binary_crossentropy",
```

```
              metrics=["accuracy"])

history = model.fit(X, Y,
                    epochs=400,
                    batch_size=128)
```

```
Epoch 1/400
 1/11             1s 165ms/step -
accuracy: 0.4375 - loss: 1.2613
```

```
/opt/anaconda3/lib/python3.12/site-packages/keras/src/layers/core/dense.py:106:
UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When
using Sequential models, prefer using an `Input(shape)` object as the first
layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
11/11             0s 2ms/step -
accuracy: 0.5831 - loss: 0.8159
Epoch 2/400
Epoch 2/400
11/11             0s 3ms/step -
accuracy: 0.6633 - loss: 0.5673
Epoch 3/400
11/11             0s 1ms/step -
accuracy: 0.7223 - loss: 0.5252
Epoch 4/400
11/11             0s 2ms/step -
accuracy: 0.7332 - loss: 0.5183
Epoch 5/400
11/11             0s 2ms/step -
accuracy: 0.7347 - loss: 0.5178
Epoch 6/400
11/11             0s 2ms/step -
accuracy: 0.7289 - loss: 0.5164
Epoch 7/400
11/11             0s 2ms/step -
accuracy: 0.7230 - loss: 0.5133
Epoch 8/400
11/11             0s 2ms/step -
accuracy: 0.7172 - loss: 0.5130
Epoch 9/400
11/11             0s 2ms/step -
accuracy: 0.7208 - loss: 0.5129
Epoch 10/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5130
Epoch 11/400
11/11             0s 2ms/step -
accuracy: 0.7194 - loss: 0.5129
```

```
Epoch 12/400
11/11             0s 2ms/step -
accuracy: 0.7136 - loss: 0.5153
Epoch 13/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5155
Epoch 14/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5188
Epoch 15/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5119
Epoch 16/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5146
Epoch 17/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5136
Epoch 18/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5119
Epoch 19/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5140
Epoch 20/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5134
Epoch 21/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5132
Epoch 22/400
11/11             0s 2ms/step -
accuracy: 0.7121 - loss: 0.5125
Epoch 23/400
11/11             0s 1ms/step -
accuracy: 0.7077 - loss: 0.5148
Epoch 24/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5127
Epoch 25/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5131
Epoch 26/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5129
Epoch 27/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5143
```

```
Epoch 28/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5135
Epoch 29/400
11/11             0s 2ms/step -
accuracy: 0.7128 - loss: 0.5127
Epoch 30/400
11/11             0s 1ms/step -
accuracy: 0.7128 - loss: 0.5125
Epoch 31/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5179
Epoch 32/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5148
Epoch 33/400
11/11             0s 1ms/step -
accuracy: 0.7106 - loss: 0.5119
Epoch 34/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5130
Epoch 35/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5134
Epoch 36/400
11/11             0s 1ms/step -
accuracy: 0.7121 - loss: 0.5124
Epoch 37/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5128
Epoch 38/400
11/11             0s 2ms/step -
accuracy: 0.7121 - loss: 0.5147
Epoch 39/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5130
Epoch 40/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5144
Epoch 41/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5178
Epoch 42/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5147
Epoch 43/400
11/11             0s 2ms/step -
accuracy: 0.7121 - loss: 0.5131
```

```
Epoch 44/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5142
Epoch 45/400
11/11             0s 2ms/step -
accuracy: 0.7034 - loss: 0.5138
Epoch 46/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5147
Epoch 47/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5137
Epoch 48/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5122
Epoch 49/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5152
Epoch 50/400
11/11             0s 2ms/step -
accuracy: 0.7121 - loss: 0.5133
Epoch 51/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5125
Epoch 52/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5129
Epoch 53/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5128
Epoch 54/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5122
Epoch 55/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5166
Epoch 56/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5171
Epoch 57/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5124
Epoch 58/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5131
Epoch 59/400
11/11             0s 2ms/step -
accuracy: 0.7121 - loss: 0.5140
```

```
Epoch 60/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5140
Epoch 61/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5119
Epoch 62/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5132
Epoch 63/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5149
Epoch 64/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5151
Epoch 65/400
11/11             0s 2ms/step -
accuracy: 0.7128 - loss: 0.5152
Epoch 66/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5125
Epoch 67/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5123
Epoch 68/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5124
Epoch 69/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5131
Epoch 70/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5138
Epoch 71/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5128
Epoch 72/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5142
Epoch 73/400
11/11             0s 2ms/step -
accuracy: 0.7048 - loss: 0.5141
Epoch 74/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5133
Epoch 75/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5122
```

```
Epoch 76/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5141
Epoch 77/400
11/11             0s 2ms/step -
accuracy: 0.7143 - loss: 0.5137
Epoch 78/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5131
Epoch 79/400
11/11             0s 2ms/step -
accuracy: 0.7048 - loss: 0.5124
Epoch 80/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5124
Epoch 81/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5130
Epoch 82/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5134
Epoch 83/400
11/11             0s 4ms/step -
accuracy: 0.7136 - loss: 0.5131
Epoch 84/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5116
Epoch 85/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5138
Epoch 86/400
11/11             0s 2ms/step -
accuracy: 0.7136 - loss: 0.5136
Epoch 87/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5130
Epoch 88/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5139
Epoch 89/400
11/11             0s 4ms/step -
accuracy: 0.7070 - loss: 0.5147
Epoch 90/400
11/11             0s 1ms/step -
accuracy: 0.7092 - loss: 0.5137
Epoch 91/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5144
```

```
Epoch 92/400
11/11             0s 2ms/step -
accuracy: 0.7048 - loss: 0.5132
Epoch 93/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5149
Epoch 94/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5134
Epoch 95/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5132
Epoch 96/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5143
Epoch 97/400
11/11             0s 1ms/step -
accuracy: 0.7070 - loss: 0.5138
Epoch 98/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5134
Epoch 99/400
11/11             0s 2ms/step -
accuracy: 0.7048 - loss: 0.5134
Epoch 100/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5127
Epoch 101/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5126
Epoch 102/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5132
Epoch 103/400
11/11             0s 1ms/step -
accuracy: 0.7048 - loss: 0.5122
Epoch 104/400
11/11             0s 2ms/step -
accuracy: 0.7048 - loss: 0.5177
Epoch 105/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5167
Epoch 106/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5158
Epoch 107/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5136
```

```
Epoch 108/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5141
Epoch 109/400
11/11             0s 2ms/step -
accuracy: 0.7157 - loss: 0.5124
Epoch 110/400
11/11             0s 1ms/step -
accuracy: 0.7092 - loss: 0.5121
Epoch 111/400
11/11             0s 2ms/step -
accuracy: 0.7121 - loss: 0.5132
Epoch 112/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5144
Epoch 113/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5127
Epoch 114/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5146
Epoch 115/400
11/11             0s 1ms/step -
accuracy: 0.7048 - loss: 0.5145
Epoch 116/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5135
Epoch 117/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5139
Epoch 118/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5140
Epoch 119/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5140
Epoch 120/400
11/11             0s 3ms/step -
accuracy: 0.7114 - loss: 0.5128
Epoch 121/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5130
Epoch 122/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5148
Epoch 123/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5136
```

```
Epoch 124/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5164
Epoch 125/400
11/11             0s 2ms/step -
accuracy: 0.7121 - loss: 0.5144
Epoch 126/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5141
Epoch 127/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5136
Epoch 128/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5129
Epoch 129/400
11/11             0s 2ms/step -
accuracy: 0.7041 - loss: 0.5137
Epoch 130/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5133
Epoch 131/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5133
Epoch 132/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5180
Epoch 133/400
11/11             0s 1ms/step -
accuracy: 0.7121 - loss: 0.5126
Epoch 134/400
11/11             0s 2ms/step -
accuracy: 0.7048 - loss: 0.5143
Epoch 135/400
11/11             0s 2ms/step -
accuracy: 0.7128 - loss: 0.5128
Epoch 136/400
11/11             0s 2ms/step -
accuracy: 0.7128 - loss: 0.5137
Epoch 137/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5137
Epoch 138/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5149
Epoch 139/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5137
```

```
Epoch 140/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5134
Epoch 141/400
11/11             0s 3ms/step -
accuracy: 0.7055 - loss: 0.5137
Epoch 142/400
11/11             0s 2ms/step -
accuracy: 0.7128 - loss: 0.5158
Epoch 143/400
11/11             0s 1ms/step -
accuracy: 0.7121 - loss: 0.5158
Epoch 144/400
11/11             0s 1ms/step -
accuracy: 0.7092 - loss: 0.5126
Epoch 145/400
11/11             0s 2ms/step -
accuracy: 0.7150 - loss: 0.5121
Epoch 146/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5131
Epoch 147/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5131
Epoch 148/400
11/11             0s 2ms/step -
accuracy: 0.7121 - loss: 0.5126
Epoch 149/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5132
Epoch 150/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5137
Epoch 151/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5149
Epoch 152/400
11/11             0s 1ms/step -
accuracy: 0.7063 - loss: 0.5135
Epoch 153/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5129
Epoch 154/400
11/11             0s 2ms/step -
accuracy: 0.7128 - loss: 0.5144
Epoch 155/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5114
```

```
Epoch 156/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5138
Epoch 157/400
11/11             0s 3ms/step -
accuracy: 0.7077 - loss: 0.5145
Epoch 158/400
11/11             0s 2ms/step -
accuracy: 0.7048 - loss: 0.5131
Epoch 159/400
11/11             0s 1ms/step -
accuracy: 0.7121 - loss: 0.5127
Epoch 160/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5135
Epoch 161/400
11/11             0s 1ms/step -
accuracy: 0.7055 - loss: 0.5127
Epoch 162/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5145
Epoch 163/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5135
Epoch 164/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5122
Epoch 165/400
11/11             0s 1ms/step -
accuracy: 0.7106 - loss: 0.5123
Epoch 166/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5160
Epoch 167/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5131
Epoch 168/400
11/11             0s 2ms/step -
accuracy: 0.7026 - loss: 0.5134
Epoch 169/400
11/11             0s 1ms/step -
accuracy: 0.7114 - loss: 0.5129
Epoch 170/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5145
Epoch 171/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5140
```

```
Epoch 172/400
11/11             0s 1ms/step -
accuracy: 0.7063 - loss: 0.5140
Epoch 173/400
11/11             0s 2ms/step -
accuracy: 0.7034 - loss: 0.5150
Epoch 174/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5143
Epoch 175/400
11/11             0s 1ms/step -
accuracy: 0.7055 - loss: 0.5132
Epoch 176/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5136
Epoch 177/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5131
Epoch 178/400
11/11             0s 2ms/step -
accuracy: 0.7048 - loss: 0.5151
Epoch 179/400
11/11             0s 1ms/step -
accuracy: 0.7150 - loss: 0.5146
Epoch 180/400
11/11             0s 1ms/step -
accuracy: 0.7121 - loss: 0.5128
Epoch 181/400
11/11             0s 2ms/step -
accuracy: 0.7136 - loss: 0.5132
Epoch 182/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5139
Epoch 183/400
11/11             0s 1ms/step -
accuracy: 0.7128 - loss: 0.5132
Epoch 184/400
11/11             0s 1ms/step -
accuracy: 0.7106 - loss: 0.5122
Epoch 185/400
11/11             0s 2ms/step -
accuracy: 0.7048 - loss: 0.5129
Epoch 186/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5124
Epoch 187/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5136
```

```
Epoch 188/400
11/11             0s 1ms/step -
accuracy: 0.7114 - loss: 0.5138
Epoch 189/400
11/11             0s 1ms/step -
accuracy: 0.7121 - loss: 0.5119
Epoch 190/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5126
Epoch 191/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5151
Epoch 192/400
11/11             0s 1ms/step -
accuracy: 0.7063 - loss: 0.5131
Epoch 193/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5136
Epoch 194/400
11/11             0s 1ms/step -
accuracy: 0.7106 - loss: 0.5141
Epoch 195/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5128
Epoch 196/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5120
Epoch 197/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5121
Epoch 198/400
11/11             0s 1ms/step -
accuracy: 0.7085 - loss: 0.5145
Epoch 199/400
11/11             0s 1ms/step -
accuracy: 0.7055 - loss: 0.5159
Epoch 200/400
11/11             0s 1ms/step -
accuracy: 0.7063 - loss: 0.5148
Epoch 201/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5131
Epoch 202/400
11/11             0s 1ms/step -
accuracy: 0.7085 - loss: 0.5126
Epoch 203/400
11/11             0s 1ms/step -
accuracy: 0.7092 - loss: 0.5129
```

```
Epoch 204/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5131
Epoch 205/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5136
Epoch 206/400
11/11             0s 3ms/step -
accuracy: 0.7048 - loss: 0.5129
Epoch 207/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5131
Epoch 208/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5134
Epoch 209/400
11/11             0s 2ms/step -
accuracy: 0.7048 - loss: 0.5164
Epoch 210/400
11/11             0s 1ms/step -
accuracy: 0.7048 - loss: 0.5121
Epoch 211/400
11/11             0s 2ms/step -
accuracy: 0.7121 - loss: 0.5140
Epoch 212/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5146
Epoch 213/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5145
Epoch 214/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5115
Epoch 215/400
11/11             0s 1ms/step -
accuracy: 0.7106 - loss: 0.5143
Epoch 216/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5172
Epoch 217/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5141
Epoch 218/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5133
Epoch 219/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5131
```

```
Epoch 220/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5144
Epoch 221/400
11/11             0s 1ms/step -
accuracy: 0.7106 - loss: 0.5125
Epoch 222/400
11/11             0s 1ms/step -
accuracy: 0.7128 - loss: 0.5131
Epoch 223/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5127
Epoch 224/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5118
Epoch 225/400
11/11             0s 1ms/step -
accuracy: 0.7077 - loss: 0.5180
Epoch 226/400
11/11             0s 2ms/step -
accuracy: 0.7034 - loss: 0.5137
Epoch 227/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5144
Epoch 228/400
11/11             0s 1ms/step -
accuracy: 0.7114 - loss: 0.5145
Epoch 229/400
11/11             0s 1ms/step -
accuracy: 0.7048 - loss: 0.5170
Epoch 230/400
11/11             0s 3ms/step -
accuracy: 0.7099 - loss: 0.5129
Epoch 231/400
11/11             0s 2ms/step -
accuracy: 0.7121 - loss: 0.5126
Epoch 232/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5145
Epoch 233/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5143
Epoch 234/400
11/11             0s 1ms/step -
accuracy: 0.7070 - loss: 0.5133
Epoch 235/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5149
```

```
Epoch 236/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5130
Epoch 237/400
11/11             0s 1ms/step -
accuracy: 0.7099 - loss: 0.5132
Epoch 238/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5139
Epoch 239/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5145
Epoch 240/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5126
Epoch 241/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5126
Epoch 242/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5129
Epoch 243/400
11/11             0s 1ms/step -
accuracy: 0.7085 - loss: 0.5129
Epoch 244/400
11/11             0s 1ms/step -
accuracy: 0.7121 - loss: 0.5129
Epoch 245/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5141
Epoch 246/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5163
Epoch 247/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5158
Epoch 248/400
11/11             0s 2ms/step -
accuracy: 0.7143 - loss: 0.5140
Epoch 249/400
11/11             0s 1ms/step -
accuracy: 0.7092 - loss: 0.5136
Epoch 250/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5123
Epoch 251/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5155
```

```
Epoch 252/400
11/11             0s 1ms/step -
accuracy: 0.7114 - loss: 0.5130
Epoch 253/400
11/11             0s 1ms/step -
accuracy: 0.7099 - loss: 0.5143
Epoch 254/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5127
Epoch 255/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5130
Epoch 256/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5140
Epoch 257/400
11/11             0s 1ms/step -
accuracy: 0.7092 - loss: 0.5116
Epoch 258/400
11/11             0s 1ms/step -
accuracy: 0.7092 - loss: 0.5132
Epoch 259/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5125
Epoch 260/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5135
Epoch 261/400
11/11             0s 1ms/step -
accuracy: 0.7070 - loss: 0.5129
Epoch 262/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5126
Epoch 263/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5185
Epoch 264/400
11/11             0s 1ms/step -
accuracy: 0.7077 - loss: 0.5135
Epoch 265/400
11/11             0s 1ms/step -
accuracy: 0.7121 - loss: 0.5126
Epoch 266/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5119
Epoch 267/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5149
```

```
Epoch 268/400
11/11             0s 1ms/step -
accuracy: 0.7070 - loss: 0.5135
Epoch 269/400
11/11             0s 1ms/step -
accuracy: 0.7092 - loss: 0.5138
Epoch 270/400
11/11             0s 2ms/step -
accuracy: 0.7121 - loss: 0.5133
Epoch 271/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5140
Epoch 272/400
11/11             0s 1ms/step -
accuracy: 0.7150 - loss: 0.5136
Epoch 273/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5138
Epoch 274/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5139
Epoch 275/400
11/11             0s 3ms/step -
accuracy: 0.7114 - loss: 0.5123
Epoch 276/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5134
Epoch 277/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5126
Epoch 278/400
11/11             0s 2ms/step -
accuracy: 0.7128 - loss: 0.5146
Epoch 279/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5122
Epoch 280/400
11/11             0s 2ms/step -
accuracy: 0.7136 - loss: 0.5145
Epoch 281/400
11/11             0s 2ms/step -
accuracy: 0.7034 - loss: 0.5128
Epoch 282/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5122
Epoch 283/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5129
```

```
Epoch 284/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5135
Epoch 285/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5136
Epoch 286/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5164
Epoch 287/400
11/11             0s 2ms/step -
accuracy: 0.7136 - loss: 0.5138
Epoch 288/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5143
Epoch 289/400
11/11             0s 2ms/step -
accuracy: 0.7048 - loss: 0.5126
Epoch 290/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5139
Epoch 291/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5129
Epoch 292/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5158
Epoch 293/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5139
Epoch 294/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5160
Epoch 295/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5145
Epoch 296/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5119
Epoch 297/400
11/11             0s 2ms/step -
accuracy: 0.7041 - loss: 0.5139
Epoch 298/400
11/11             0s 2ms/step -
accuracy: 0.7143 - loss: 0.5122
Epoch 299/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5136
```

```
Epoch 300/400
11/11             0s 2ms/step -
accuracy: 0.7034 - loss: 0.5140
Epoch 301/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5151
Epoch 302/400
11/11             0s 2ms/step -
accuracy: 0.7121 - loss: 0.5136
Epoch 303/400
11/11             0s 2ms/step -
accuracy: 0.7128 - loss: 0.5138
Epoch 304/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5164
Epoch 305/400
11/11             0s 2ms/step -
accuracy: 0.7136 - loss: 0.5140
Epoch 306/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5140
Epoch 307/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5141
Epoch 308/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5137
Epoch 309/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5132
Epoch 310/400
11/11             0s 3ms/step -
accuracy: 0.7085 - loss: 0.5130
Epoch 311/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5118
Epoch 312/400
11/11             0s 2ms/step -
accuracy: 0.7026 - loss: 0.5143
Epoch 313/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5139
Epoch 314/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5134
Epoch 315/400
11/11             0s 3ms/step -
accuracy: 0.7085 - loss: 0.5141
```

```
Epoch 316/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5114
Epoch 317/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5131
Epoch 318/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5146
Epoch 319/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5137
Epoch 320/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5122
Epoch 321/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5160
Epoch 322/400
11/11             0s 2ms/step -
accuracy: 0.7004 - loss: 0.5168
Epoch 323/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5150
Epoch 324/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5109
Epoch 325/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5133
Epoch 326/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5123
Epoch 327/400
11/11             0s 2ms/step -
accuracy: 0.7019 - loss: 0.5130
Epoch 328/400
11/11             0s 2ms/step -
accuracy: 0.7128 - loss: 0.5149
Epoch 329/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5145
Epoch 330/400
11/11             0s 2ms/step -
accuracy: 0.7048 - loss: 0.5151
Epoch 331/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5125
```

```
Epoch 332/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5127
Epoch 333/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5136
Epoch 334/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5141
Epoch 335/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5120
Epoch 336/400
11/11             0s 2ms/step -
accuracy: 0.7136 - loss: 0.5118
Epoch 337/400
11/11             0s 2ms/step -
accuracy: 0.7048 - loss: 0.5132
Epoch 338/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5133
Epoch 339/400
11/11             0s 2ms/step -
accuracy: 0.7041 - loss: 0.5137
Epoch 340/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5152
Epoch 341/400
11/11             0s 2ms/step -
accuracy: 0.7128 - loss: 0.5154
Epoch 342/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5135
Epoch 343/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5164
Epoch 344/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5124
Epoch 345/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5122
Epoch 346/400
11/11             0s 3ms/step -
accuracy: 0.7099 - loss: 0.5154
Epoch 347/400
11/11             0s 2ms/step -
accuracy: 0.7041 - loss: 0.5119
```

```
Epoch 348/400
11/11             0s 2ms/step -
accuracy: 0.7114 - loss: 0.5138
Epoch 349/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5128
Epoch 350/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5121
Epoch 351/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5140
Epoch 352/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5136
Epoch 353/400
11/11             0s 1ms/step -
accuracy: 0.7041 - loss: 0.5136
Epoch 354/400
11/11             0s 2ms/step -
accuracy: 0.7034 - loss: 0.5159
Epoch 355/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5146
Epoch 356/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5131
Epoch 357/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5141
Epoch 358/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5123
Epoch 359/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5124
Epoch 360/400
11/11             0s 1ms/step -
accuracy: 0.7092 - loss: 0.5128
Epoch 361/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5141
Epoch 362/400
11/11             0s 1ms/step -
accuracy: 0.7063 - loss: 0.5128
Epoch 363/400
11/11             0s 3ms/step -
accuracy: 0.7136 - loss: 0.5127
```

```
Epoch 364/400
11/11             0s 2ms/step -
accuracy: 0.7048 - loss: 0.5160
Epoch 365/400
11/11             0s 1ms/step -
accuracy: 0.7106 - loss: 0.5135
Epoch 366/400
11/11             0s 2ms/step -
accuracy: 0.7092 - loss: 0.5137
Epoch 367/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5152
Epoch 368/400
11/11             0s 2ms/step -
accuracy: 0.7121 - loss: 0.5143
Epoch 369/400
11/11             0s 2ms/step -
accuracy: 0.7128 - loss: 0.5158
Epoch 370/400
11/11             0s 1ms/step -
accuracy: 0.7099 - loss: 0.5148
Epoch 371/400
11/11             0s 2ms/step -
accuracy: 0.7128 - loss: 0.5124
Epoch 372/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5139
Epoch 373/400
11/11             0s 1ms/step -
accuracy: 0.7048 - loss: 0.5131
Epoch 374/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5137
Epoch 375/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5136
Epoch 376/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5151
Epoch 377/400
11/11             0s 2ms/step -
accuracy: 0.7106 - loss: 0.5149
Epoch 378/400
11/11             0s 3ms/step -
accuracy: 0.7099 - loss: 0.5126
Epoch 379/400
11/11             0s 2ms/step -
accuracy: 0.7055 - loss: 0.5137
```

```
Epoch 380/400
11/11             0s 1ms/step -
accuracy: 0.7092 - loss: 0.5127
Epoch 381/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5130
Epoch 382/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5139
Epoch 383/400
11/11             0s 1ms/step -
accuracy: 0.7077 - loss: 0.5123
Epoch 384/400
11/11             0s 1ms/step -
accuracy: 0.7063 - loss: 0.5145
Epoch 385/400
11/11             0s 1ms/step -
accuracy: 0.7092 - loss: 0.5114
Epoch 386/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5136
Epoch 387/400
11/11             0s 4ms/step -
accuracy: 0.7092 - loss: 0.5136
Epoch 388/400
11/11             0s 1ms/step -
accuracy: 0.7041 - loss: 0.5167
Epoch 389/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5147
Epoch 390/400
11/11             0s 2ms/step -
accuracy: 0.7143 - loss: 0.5132
Epoch 391/400
11/11             0s 2ms/step -
accuracy: 0.7150 - loss: 0.5125
Epoch 392/400
11/11             0s 2ms/step -
accuracy: 0.7048 - loss: 0.5132
Epoch 393/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5164
Epoch 394/400
11/11             0s 1ms/step -
accuracy: 0.7055 - loss: 0.5138
Epoch 395/400
11/11             0s 3ms/step -
accuracy: 0.7143 - loss: 0.5118
```

```
Epoch 396/400
11/11             0s 2ms/step -
accuracy: 0.7070 - loss: 0.5123
Epoch 397/400
11/11             0s 2ms/step -
accuracy: 0.7034 - loss: 0.5131
Epoch 398/400
11/11             0s 2ms/step -
accuracy: 0.7063 - loss: 0.5147
Epoch 399/400
11/11             0s 2ms/step -
accuracy: 0.7128 - loss: 0.5125
Epoch 400/400
11/11             0s 2ms/step -
accuracy: 0.7085 - loss: 0.5137
```

[30]:
```
# Visualisation
plt.plot(history.history["loss"])
plt.title("Training loss")
plt.xlabel("Epoch")
plt.ylabel("Binary crossentropy")
plt.show()

plt.plot(history.history["accuracy"])
plt.title("Training accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.show()
```

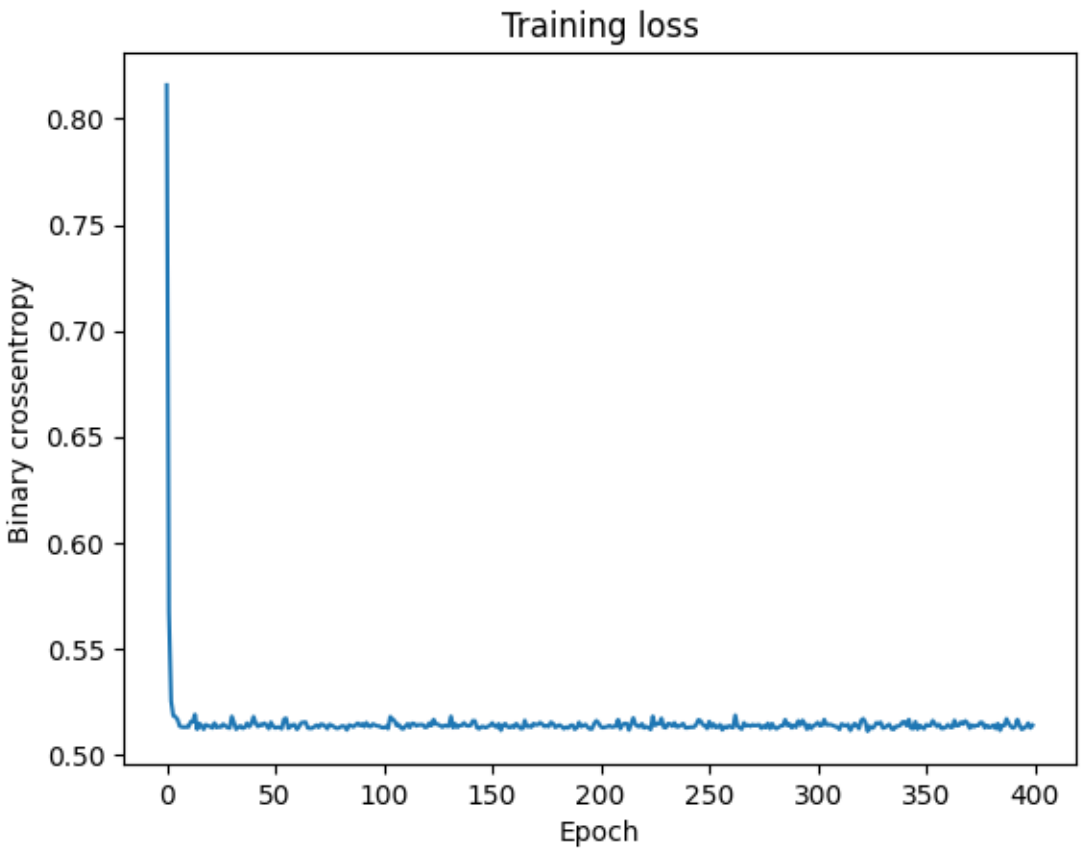
```
<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

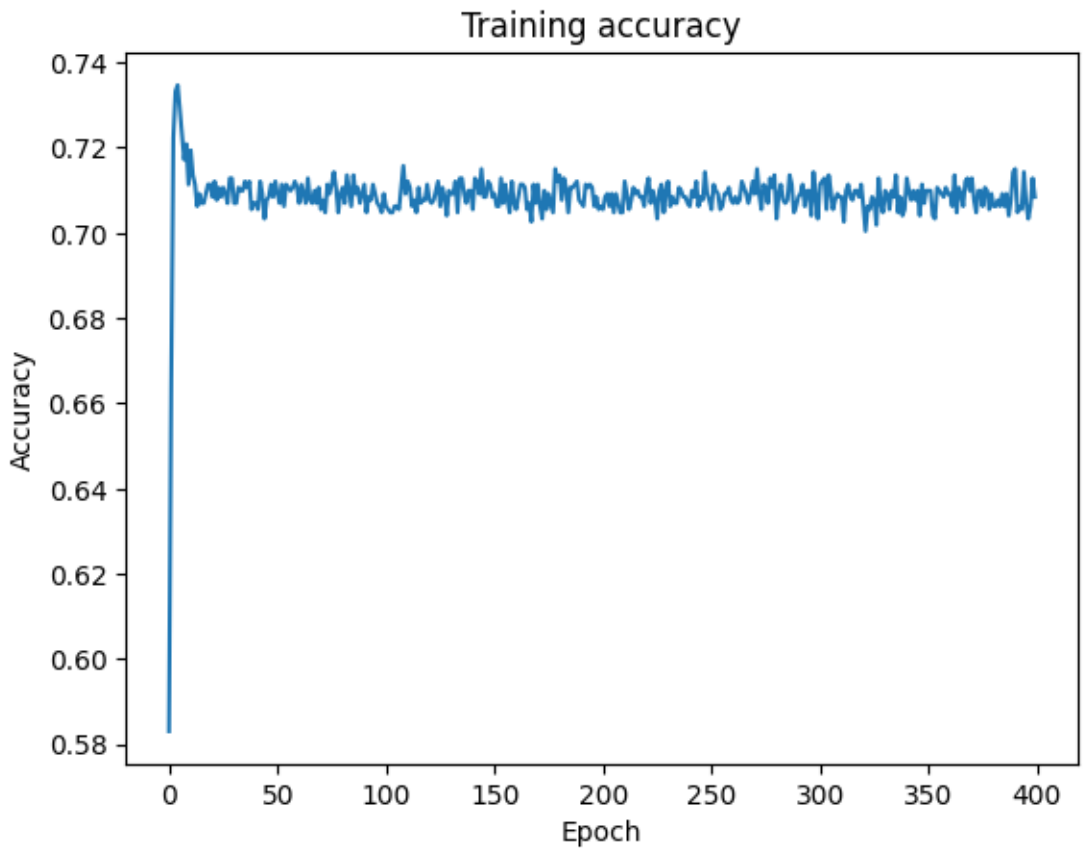
<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>
```




```
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
```



The logistic regression model reached an accuracy of 0.70 on the training set, which is relatively poor for a binary classification task a random classifier would already achieve 0.50. The data is not linearly separable in the original feature space.

1.3 6.3

[31]:
```
x1list = np.linspace(np.min(X[:,0])-2, np.max(X[:,0])+2, 50)
x2list = np.linspace(np.min(X[:,1])-2, np.max(X[:,1])+2, 50)
```

[36]:
```
x1grid, x2grid = np.meshgrid(x1list, x2list)

grid_points = np.c_[x1grid.ravel(), x2grid.ravel()]
Z = model.predict(grid_points)
Z = Z.reshape(x1grid.shape)

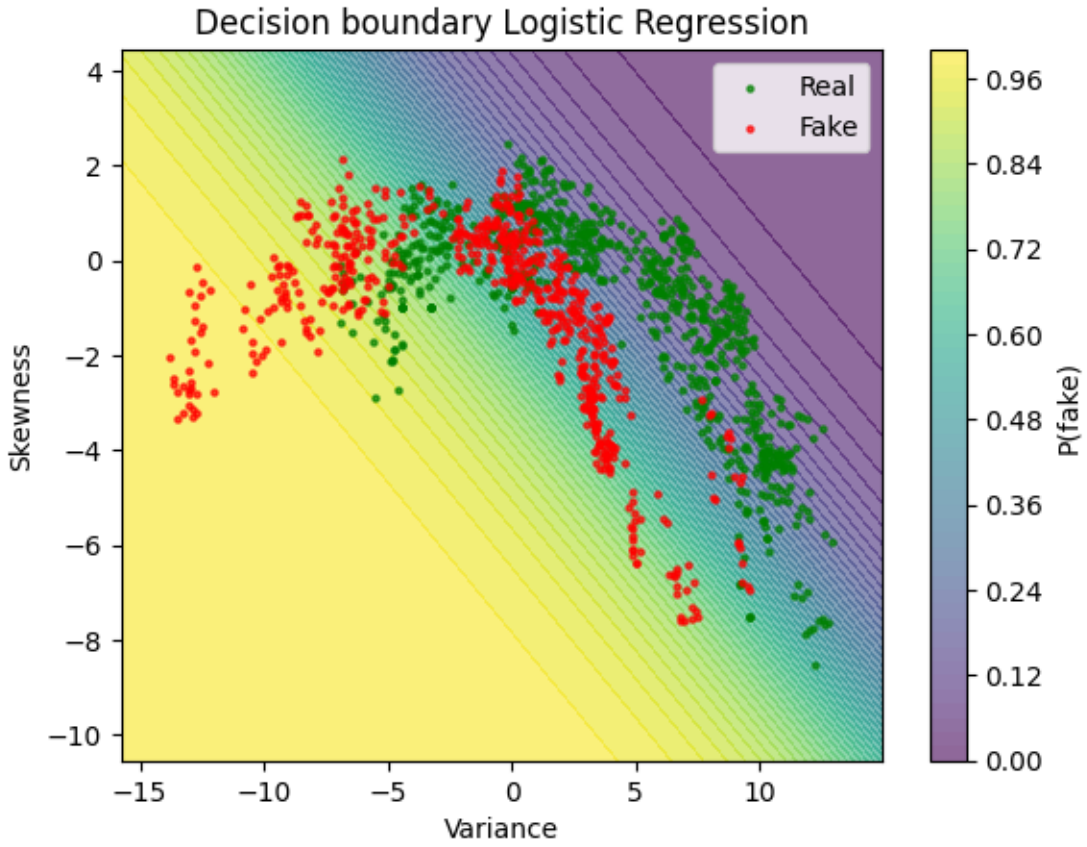
plt.contourf(x1grid, x2grid, Z, levels=50, cmap="viridis", alpha=0.6)
plt.colorbar(label="P(fake)")

plt.scatter(real[:, 0], real[:, 1], color="green", s=5, alpha=0.7, label="Real")
plt.scatter(fake[:, 0], fake[:, 1], color="red",   s=5, alpha=0.7, label="Fake")
```

```
plt.title("Decision boundary Logistic Regression")
plt.xlabel("Variance")
plt.ylabel("Skewness")
plt.legend()
plt.show()
```

```
79/79             0s 443us/step
```

```
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
```



1.4 6.4

[38]:
```
model = Sequential([
    Dense(1, input_shape=(X.shape[1],), activation="sigmoid"),
    Dense(8, input_shape=(X.shape[1],), activation="sigmoid"),
    Dense(2, input_shape=(X.shape[1],), activation="softmax")
])

model.compile(optimizer=SGD(learning_rate=0.15),
              loss="categorical_crossentropy",
              metrics=["accuracy"])

from tensorflow.keras.utils import to_categorical

Y_c= to_categorical(Y,2)

history = model.fit(X, Y_c,
                    epochs=400,
                    batch_size=128)
```

```
Epoch 1/400
11/11             0s 2ms/step -
accuracy: 0.5160 - loss: 0.6954
Epoch 2/400
11/11             0s 2ms/step -
accuracy: 0.5554 - loss: 0.6875
Epoch 3/400
11/11             0s 2ms/step -
accuracy: 0.5554 - loss: 0.6861
Epoch 4/400
11/11             0s 2ms/step -
accuracy: 0.5554 - loss: 0.6848
Epoch 5/400
11/11             0s 2ms/step -
accuracy: 0.5554 - loss: 0.6851
Epoch 6/400
11/11             0s 2ms/step -
accuracy: 0.5554 - loss: 0.6831
Epoch 7/400
11/11             0s 2ms/step -
accuracy: 0.5554 - loss: 0.6802
Epoch 8/400
11/11             0s 2ms/step -
accuracy: 0.5554 - loss: 0.6779
Epoch 9/400
```

```
11/11             0s 2ms/step -
accuracy: 0.5554 - loss: 0.6750
Epoch 10/400
11/11             0s 2ms/step -
accuracy: 0.5554 - loss: 0.6699
Epoch 11/400
11/11             0s 2ms/step -
accuracy: 0.5554 - loss: 0.6646
Epoch 12/400
11/11             0s 2ms/step -
accuracy: 0.5590 - loss: 0.6591
Epoch 13/400
11/11             0s 4ms/step -
accuracy: 0.5751 - loss: 0.6527
Epoch 14/400
11/11             0s 2ms/step -
accuracy: 0.6115 - loss: 0.6466
Epoch 15/400
11/11             0s 2ms/step -
accuracy: 0.6633 - loss: 0.6409
Epoch 16/400
11/11             0s 2ms/step -
accuracy: 0.6713 - loss: 0.6349
Epoch 17/400
11/11             0s 2ms/step -
accuracy: 0.6611 - loss: 0.6292
Epoch 18/400
11/11             0s 2ms/step -
accuracy: 0.6698 - loss: 0.6239
Epoch 19/400
11/11             0s 2ms/step -
accuracy: 0.7004 - loss: 0.6167
Epoch 20/400
11/11             0s 2ms/step -
accuracy: 0.6698 - loss: 0.6119
Epoch 21/400
11/11             0s 2ms/step -
accuracy: 0.6968 - loss: 0.6048
Epoch 22/400
11/11             0s 2ms/step -
accuracy: 0.6968 - loss: 0.5994
Epoch 23/400
11/11             0s 2ms/step -
accuracy: 0.6866 - loss: 0.5927
Epoch 24/400
11/11             0s 2ms/step -
accuracy: 0.7077 - loss: 0.5885
Epoch 25/400
```

```
11/11             0s 3ms/step -
accuracy: 0.6990 - loss: 0.5811
Epoch 26/400
11/11             0s 2ms/step -
accuracy: 0.7099 - loss: 0.5761
Epoch 27/400
11/11             0s 2ms/step -
accuracy: 0.7128 - loss: 0.5709
Epoch 28/400
11/11             0s 2ms/step -
accuracy: 0.7274 - loss: 0.5649
Epoch 29/400
11/11             0s 2ms/step -
accuracy: 0.7172 - loss: 0.5593
Epoch 30/400
11/11             0s 2ms/step -
accuracy: 0.7216 - loss: 0.5538
Epoch 31/400
11/11             0s 2ms/step -
accuracy: 0.7332 - loss: 0.5486
Epoch 32/400
11/11             0s 2ms/step -
accuracy: 0.7325 - loss: 0.5439
Epoch 33/400
11/11             0s 2ms/step -
accuracy: 0.7391 - loss: 0.5392
Epoch 34/400
11/11             0s 2ms/step -
accuracy: 0.7325 - loss: 0.5347
Epoch 35/400
11/11             0s 2ms/step -
accuracy: 0.7398 - loss: 0.5306
Epoch 36/400
11/11             0s 2ms/step -
accuracy: 0.7376 - loss: 0.5265
Epoch 37/400
11/11             0s 2ms/step -
accuracy: 0.7413 - loss: 0.5222
Epoch 38/400
11/11             0s 4ms/step -
accuracy: 0.7405 - loss: 0.5186
Epoch 39/400
11/11             0s 2ms/step -
accuracy: 0.7478 - loss: 0.5154
Epoch 40/400
11/11             0s 2ms/step -
accuracy: 0.7434 - loss: 0.5112
Epoch 41/400
```

```
11/11             0s 2ms/step -
accuracy: 0.7515 - loss: 0.5073
Epoch 42/400
11/11             0s 2ms/step -
accuracy: 0.7485 - loss: 0.5042
Epoch 43/400
11/11             0s 2ms/step -
accuracy: 0.7536 - loss: 0.5013
Epoch 44/400
11/11             0s 2ms/step -
accuracy: 0.7573 - loss: 0.4978
Epoch 45/400
11/11             0s 2ms/step -
accuracy: 0.7587 - loss: 0.4946
Epoch 46/400
11/11             0s 2ms/step -
accuracy: 0.7631 - loss: 0.4914
Epoch 47/400
11/11             0s 2ms/step -
accuracy: 0.7595 - loss: 0.4890
Epoch 48/400
11/11             0s 2ms/step -
accuracy: 0.7719 - loss: 0.4855
Epoch 49/400
11/11             0s 2ms/step -
accuracy: 0.7646 - loss: 0.4834
Epoch 50/400
11/11             0s 2ms/step -
accuracy: 0.7675 - loss: 0.4807
Epoch 51/400
11/11             0s 2ms/step -
accuracy: 0.7690 - loss: 0.4783
Epoch 52/400
11/11             0s 2ms/step -
accuracy: 0.7762 - loss: 0.4756
Epoch 53/400
11/11             0s 2ms/step -
accuracy: 0.7711 - loss: 0.4744
Epoch 54/400
11/11             0s 2ms/step -
accuracy: 0.7733 - loss: 0.4715
Epoch 55/400
11/11             0s 2ms/step -
accuracy: 0.7770 - loss: 0.4686
Epoch 56/400
11/11             0s 2ms/step -
accuracy: 0.7777 - loss: 0.4672
Epoch 57/400
```

```
11/11             0s 4ms/step -
accuracy: 0.7792 - loss: 0.4651
Epoch 58/400
11/11             0s 2ms/step -
accuracy: 0.7792 - loss: 0.4632
Epoch 59/400
11/11             0s 2ms/step -
accuracy: 0.7821 - loss: 0.4610
Epoch 60/400
11/11             0s 2ms/step -
accuracy: 0.7850 - loss: 0.4597
Epoch 61/400
11/11             0s 2ms/step -
accuracy: 0.7835 - loss: 0.4578
Epoch 62/400
11/11             0s 2ms/step -
accuracy: 0.7872 - loss: 0.4561
Epoch 63/400
11/11             0s 2ms/step -
accuracy: 0.7857 - loss: 0.4539
Epoch 64/400
11/11             0s 2ms/step -
accuracy: 0.7872 - loss: 0.4526
Epoch 65/400
11/11             0s 2ms/step -
accuracy: 0.7843 - loss: 0.4518
Epoch 66/400
11/11             0s 2ms/step -
accuracy: 0.7915 - loss: 0.4500
Epoch 67/400
11/11             0s 2ms/step -
accuracy: 0.7886 - loss: 0.4493
Epoch 68/400
11/11             0s 2ms/step -
accuracy: 0.7959 - loss: 0.4477
Epoch 69/400
11/11             0s 2ms/step -
accuracy: 0.7901 - loss: 0.4467
Epoch 70/400
11/11             0s 2ms/step -
accuracy: 0.7952 - loss: 0.4450
Epoch 71/400
11/11             0s 2ms/step -
accuracy: 0.7937 - loss: 0.4440
Epoch 72/400
11/11             0s 2ms/step -
accuracy: 0.7937 - loss: 0.4432
Epoch 73/400
```

```
11/11             0s 2ms/step -
accuracy: 0.7937 - loss: 0.4418
Epoch 74/400
11/11             0s 2ms/step -
accuracy: 0.7937 - loss: 0.4413
Epoch 75/400
11/11             0s 2ms/step -
accuracy: 0.7952 - loss: 0.4400
Epoch 76/400
11/11             0s 2ms/step -
accuracy: 0.8003 - loss: 0.4392
Epoch 77/400
11/11             0s 2ms/step -
accuracy: 0.7952 - loss: 0.4394
Epoch 78/400
11/11             0s 2ms/step -
accuracy: 0.7959 - loss: 0.4382
Epoch 79/400
11/11             0s 2ms/step -
accuracy: 0.7981 - loss: 0.4371
Epoch 80/400
11/11             0s 2ms/step -
accuracy: 0.8010 - loss: 0.4360
Epoch 81/400
11/11             0s 2ms/step -
accuracy: 0.7981 - loss: 0.4363
Epoch 82/400
11/11             0s 2ms/step -
accuracy: 0.8003 - loss: 0.4352
Epoch 83/400
11/11             0s 1ms/step -
accuracy: 0.7981 - loss: 0.4339
Epoch 84/400
11/11             0s 2ms/step -
accuracy: 0.8047 - loss: 0.4349
Epoch 85/400
11/11             0s 2ms/step -
accuracy: 0.7974 - loss: 0.4339
Epoch 86/400
11/11             0s 3ms/step -
accuracy: 0.7988 - loss: 0.4326
Epoch 87/400
11/11             0s 2ms/step -
accuracy: 0.8010 - loss: 0.4320
Epoch 88/400
11/11             0s 2ms/step -
accuracy: 0.8025 - loss: 0.4321
Epoch 89/400
```

```
11/11             0s 2ms/step -
accuracy: 0.7996 - loss: 0.4307
Epoch 90/400
11/11             0s 2ms/step -
accuracy: 0.8039 - loss: 0.4304
Epoch 91/400
11/11             0s 1ms/step -
accuracy: 0.8032 - loss: 0.4291
Epoch 92/400
11/11             0s 2ms/step -
accuracy: 0.7988 - loss: 0.4304
Epoch 93/400
11/11             0s 2ms/step -
accuracy: 0.8032 - loss: 0.4296
Epoch 94/400
11/11             0s 3ms/step -
accuracy: 0.8069 - loss: 0.4284
Epoch 95/400
11/11             0s 2ms/step -
accuracy: 0.8017 - loss: 0.4289
Epoch 96/400
11/11             0s 2ms/step -
accuracy: 0.8003 - loss: 0.4286
Epoch 97/400
11/11             0s 2ms/step -
accuracy: 0.8047 - loss: 0.4278
Epoch 98/400
11/11             0s 2ms/step -
accuracy: 0.8039 - loss: 0.4270
Epoch 99/400
11/11             0s 2ms/step -
accuracy: 0.8069 - loss: 0.4265
Epoch 100/400
11/11             0s 2ms/step -
accuracy: 0.8039 - loss: 0.4275
Epoch 101/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4252
Epoch 102/400
11/11             0s 2ms/step -
accuracy: 0.8003 - loss: 0.4264
Epoch 103/400
11/11             0s 2ms/step -
accuracy: 0.8054 - loss: 0.4256
Epoch 104/400
11/11             0s 2ms/step -
accuracy: 0.8025 - loss: 0.4250
Epoch 105/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8054 - loss: 0.4257
Epoch 106/400
11/11             0s 2ms/step -
accuracy: 0.8061 - loss: 0.4249
Epoch 107/400
11/11             0s 2ms/step -
accuracy: 0.8076 - loss: 0.4239
Epoch 108/400
11/11             0s 2ms/step -
accuracy: 0.8098 - loss: 0.4249
Epoch 109/400
11/11             0s 3ms/step -
accuracy: 0.8039 - loss: 0.4246
Epoch 110/400
11/11             0s 2ms/step -
accuracy: 0.8069 - loss: 0.4230
Epoch 111/400
11/11             0s 2ms/step -
accuracy: 0.8054 - loss: 0.4232
Epoch 112/400
11/11             0s 2ms/step -
accuracy: 0.8076 - loss: 0.4233
Epoch 113/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4222
Epoch 114/400
11/11             0s 2ms/step -
accuracy: 0.8076 - loss: 0.4225
Epoch 115/400
11/11             0s 2ms/step -
accuracy: 0.8112 - loss: 0.4214
Epoch 116/400
11/11             0s 2ms/step -
accuracy: 0.8090 - loss: 0.4219
Epoch 117/400
11/11             0s 2ms/step -
accuracy: 0.8054 - loss: 0.4215
Epoch 118/400
11/11             0s 2ms/step -
accuracy: 0.8090 - loss: 0.4215
Epoch 119/400
11/11             0s 2ms/step -
accuracy: 0.8098 - loss: 0.4206
Epoch 120/400
11/11             0s 2ms/step -
accuracy: 0.8090 - loss: 0.4219
Epoch 121/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8112 - loss: 0.4216
Epoch 122/400
11/11             0s 2ms/step -
accuracy: 0.8083 - loss: 0.4214
Epoch 123/400
11/11             0s 2ms/step -
accuracy: 0.8083 - loss: 0.4205
Epoch 124/400
11/11             0s 2ms/step -
accuracy: 0.8105 - loss: 0.4206
Epoch 125/400
11/11             0s 2ms/step -
accuracy: 0.8090 - loss: 0.4202
Epoch 126/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4207
Epoch 127/400
11/11             0s 2ms/step -
accuracy: 0.8098 - loss: 0.4197
Epoch 128/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4199
Epoch 129/400
11/11             0s 2ms/step -
accuracy: 0.8083 - loss: 0.4191
Epoch 130/400
11/11             0s 2ms/step -
accuracy: 0.8112 - loss: 0.4195
Epoch 131/400
11/11             0s 2ms/step -
accuracy: 0.8105 - loss: 0.4200
Epoch 132/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4194
Epoch 133/400
11/11             0s 3ms/step -
accuracy: 0.8112 - loss: 0.4189
Epoch 134/400
11/11             0s 2ms/step -
accuracy: 0.8112 - loss: 0.4187
Epoch 135/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4192
Epoch 136/400
11/11             0s 2ms/step -
accuracy: 0.8112 - loss: 0.4181
Epoch 137/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4182
Epoch 138/400
11/11             0s 4ms/step -
accuracy: 0.8105 - loss: 0.4180
Epoch 139/400
11/11             0s 2ms/step -
accuracy: 0.8112 - loss: 0.4181
Epoch 140/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4187
Epoch 141/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4185
Epoch 142/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4179
Epoch 143/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4179
Epoch 144/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4179
Epoch 145/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4178
Epoch 146/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4177
Epoch 147/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4180
Epoch 148/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4171
Epoch 149/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4175
Epoch 150/400
11/11             0s 2ms/step -
accuracy: 0.8112 - loss: 0.4180
Epoch 151/400
11/11             0s 2ms/step -
accuracy: 0.8112 - loss: 0.4172
Epoch 152/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4162
Epoch 153/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8098 - loss: 0.4166
Epoch 154/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4174
Epoch 155/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4166
Epoch 156/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4166
Epoch 157/400
11/11             0s 2ms/step -
accuracy: 0.8098 - loss: 0.4167
Epoch 158/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4160
Epoch 159/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4164
Epoch 160/400
11/11             0s 2ms/step -
accuracy: 0.8112 - loss: 0.4160
Epoch 161/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4163
Epoch 162/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4162
Epoch 163/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4163
Epoch 164/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4163
Epoch 165/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4163
Epoch 166/400
11/11             0s 4ms/step -
accuracy: 0.8141 - loss: 0.4165
Epoch 167/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4154
Epoch 168/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4161
Epoch 169/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4165
Epoch 170/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4160
Epoch 171/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4161
Epoch 172/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4155
Epoch 173/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4150
Epoch 174/400
11/11             0s 2ms/step -
accuracy: 0.8163 - loss: 0.4145
Epoch 175/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4156
Epoch 176/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4153
Epoch 177/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4157
Epoch 178/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4154
Epoch 179/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4153
Epoch 180/400
11/11             0s 2ms/step -
accuracy: 0.8105 - loss: 0.4165
Epoch 181/400
11/11             0s 1ms/step -
accuracy: 0.8134 - loss: 0.4147
Epoch 182/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4148
Epoch 183/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4152
Epoch 184/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4148
Epoch 185/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4150
Epoch 186/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4147
Epoch 187/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4144
Epoch 188/400
11/11             0s 5ms/step -
accuracy: 0.8127 - loss: 0.4144
Epoch 189/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4150
Epoch 190/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4138
Epoch 191/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4146
Epoch 192/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4145
Epoch 193/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4144
Epoch 194/400
11/11             0s 2ms/step -
accuracy: 0.8112 - loss: 0.4148
Epoch 195/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4149
Epoch 196/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4143
Epoch 197/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4142
Epoch 198/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4153
Epoch 199/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4155
Epoch 200/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4145
Epoch 201/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4134
Epoch 202/400
11/11             0s 3ms/step -
accuracy: 0.8134 - loss: 0.4149
Epoch 203/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4135
Epoch 204/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4140
Epoch 205/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4137
Epoch 206/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4148
Epoch 207/400
11/11             0s 2ms/step -
accuracy: 0.8112 - loss: 0.4145
Epoch 208/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4137
Epoch 209/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4140
Epoch 210/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4159
Epoch 211/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4145
Epoch 212/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4137
Epoch 213/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4135
Epoch 214/400
11/11             0s 3ms/step -
accuracy: 0.8149 - loss: 0.4132
Epoch 215/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4133
Epoch 216/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4133
Epoch 217/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4135
Epoch 218/400
11/11             0s 2ms/step -
accuracy: 0.8112 - loss: 0.4143
Epoch 219/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4132
Epoch 220/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4148
Epoch 221/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4135
Epoch 222/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4140
Epoch 223/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4133
Epoch 224/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4157
Epoch 225/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4141
Epoch 226/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4136
Epoch 227/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4136
Epoch 228/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4140
Epoch 229/400
11/11             0s 2ms/step -
accuracy: 0.8163 - loss: 0.4139
Epoch 230/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4137
Epoch 231/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4148
Epoch 232/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4141
Epoch 233/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4130
Epoch 234/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4140
Epoch 235/400
11/11             0s 4ms/step -
accuracy: 0.8120 - loss: 0.4134
Epoch 236/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4131
Epoch 237/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4140
Epoch 238/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4132
Epoch 239/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4132
Epoch 240/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4133
Epoch 241/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4130
Epoch 242/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4142
Epoch 243/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4131
Epoch 244/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4132
Epoch 245/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4142
Epoch 246/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4132
Epoch 247/400
11/11             0s 2ms/step -
accuracy: 0.8090 - loss: 0.4147
Epoch 248/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4143
Epoch 249/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8105 - loss: 0.4150
Epoch 250/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4131
Epoch 251/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4131
Epoch 252/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4129
Epoch 253/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4130
Epoch 254/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4144
Epoch 255/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4138
Epoch 256/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4138
Epoch 257/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4128
Epoch 258/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4128
Epoch 259/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4123
Epoch 260/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4129
Epoch 261/400
11/11             0s 2ms/step -
accuracy: 0.8105 - loss: 0.4133
Epoch 262/400
11/11             0s 3ms/step -
accuracy: 0.8134 - loss: 0.4127
Epoch 263/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4125
Epoch 264/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4136
Epoch 265/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4128
Epoch 266/400
11/11             0s 2ms/step -
accuracy: 0.8171 - loss: 0.4133
Epoch 267/400
11/11             0s 2ms/step -
accuracy: 0.8171 - loss: 0.4131
Epoch 268/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4124
Epoch 269/400
11/11             0s 2ms/step -
accuracy: 0.8098 - loss: 0.4134
Epoch 270/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4123
Epoch 271/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4130
Epoch 272/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4127
Epoch 273/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4127
Epoch 274/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4127
Epoch 275/400
11/11             0s 4ms/step -
accuracy: 0.8163 - loss: 0.4136
Epoch 276/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4125
Epoch 277/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4125
Epoch 278/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4136
Epoch 279/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4136
Epoch 280/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4133
Epoch 281/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4124
Epoch 282/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4128
Epoch 283/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4132
Epoch 284/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4133
Epoch 285/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4127
Epoch 286/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4124
Epoch 287/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4120
Epoch 288/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4123
Epoch 289/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4124
Epoch 290/400
11/11             0s 1ms/step -
accuracy: 0.8171 - loss: 0.4125
Epoch 291/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4127
Epoch 292/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4129
Epoch 293/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4122
Epoch 294/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4124
Epoch 295/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4132
Epoch 296/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4124
Epoch 297/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8171 - loss: 0.4124
Epoch 298/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4131
Epoch 299/400
11/11             0s 3ms/step -
accuracy: 0.8112 - loss: 0.4139
Epoch 300/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4139
Epoch 301/400
11/11             0s 3ms/step -
accuracy: 0.8141 - loss: 0.4127
Epoch 302/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4127
Epoch 303/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4123
Epoch 304/400
11/11             0s 2ms/step -
accuracy: 0.8163 - loss: 0.4125
Epoch 305/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4130
Epoch 306/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4123
Epoch 307/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4130
Epoch 308/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4126
Epoch 309/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4122
Epoch 310/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4119
Epoch 311/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4121
Epoch 312/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4123
Epoch 313/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4121
Epoch 314/400
11/11             0s 2ms/step -
accuracy: 0.8171 - loss: 0.4125
Epoch 315/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4128
Epoch 316/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4124
Epoch 317/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4126
Epoch 318/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4125
Epoch 319/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4126
Epoch 320/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4120
Epoch 321/400
11/11             0s 2ms/step -
accuracy: 0.8163 - loss: 0.4127
Epoch 322/400
11/11             0s 3ms/step -
accuracy: 0.8141 - loss: 0.4123
Epoch 323/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4124
Epoch 324/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4123
Epoch 325/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4134
Epoch 326/400
11/11             0s 2ms/step -
accuracy: 0.8171 - loss: 0.4121
Epoch 327/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4117
Epoch 328/400
11/11             0s 2ms/step -
accuracy: 0.8112 - loss: 0.4129
Epoch 329/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4123
Epoch 330/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4119
Epoch 331/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4118
Epoch 332/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4119
Epoch 333/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4118
Epoch 334/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4121
Epoch 335/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4120
Epoch 336/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4123
Epoch 337/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4127
Epoch 338/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4125
Epoch 339/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4128
Epoch 340/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4117
Epoch 341/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4114
Epoch 342/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4120
Epoch 343/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4122
Epoch 344/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4119
Epoch 345/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4123
Epoch 346/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4125
Epoch 347/400
11/11             0s 7ms/step -
accuracy: 0.8163 - loss: 0.4116
Epoch 348/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4117
Epoch 349/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4119
Epoch 350/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4116
Epoch 351/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4116
Epoch 352/400
11/11             0s 2ms/step -
accuracy: 0.8083 - loss: 0.4128
Epoch 353/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4121
Epoch 354/400
11/11             0s 2ms/step -
accuracy: 0.8163 - loss: 0.4125
Epoch 355/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4126
Epoch 356/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4136
Epoch 357/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4119
Epoch 358/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4121
Epoch 359/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4120
Epoch 360/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4123
Epoch 361/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4122
Epoch 362/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4122
Epoch 363/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4119
Epoch 364/400
11/11             0s 4ms/step -
accuracy: 0.8127 - loss: 0.4121
Epoch 365/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4112
Epoch 366/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4128
Epoch 367/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4120
Epoch 368/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4127
Epoch 369/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4112
Epoch 370/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4130
Epoch 371/400
11/11             0s 1ms/step -
accuracy: 0.8134 - loss: 0.4123
Epoch 372/400
11/11             0s 3ms/step -
accuracy: 0.8156 - loss: 0.4116
Epoch 373/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4116
Epoch 374/400
11/11             0s 2ms/step -
accuracy: 0.8156 - loss: 0.4121
Epoch 375/400
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4130
Epoch 376/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4114
Epoch 377/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8127 - loss: 0.4120
Epoch 378/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4121
Epoch 379/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4122
Epoch 380/400
11/11             0s 2ms/step -
accuracy: 0.8149 - loss: 0.4115
Epoch 381/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4118
Epoch 382/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4115
Epoch 383/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4118
Epoch 384/400
11/11             0s 3ms/step -
accuracy: 0.8156 - loss: 0.4126
Epoch 385/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4124
Epoch 386/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4116
Epoch 387/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4113
Epoch 388/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4126
Epoch 389/400
11/11             0s 3ms/step -
accuracy: 0.8127 - loss: 0.4115
Epoch 390/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4119
Epoch 391/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4120
Epoch 392/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4118
Epoch 393/400
```

```
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4122
Epoch 394/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4117
Epoch 395/400
11/11             0s 2ms/step -
accuracy: 0.8120 - loss: 0.4118
Epoch 396/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4126
Epoch 397/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4122
Epoch 398/400
11/11             0s 2ms/step -
accuracy: 0.8134 - loss: 0.4121
Epoch 399/400
11/11             0s 2ms/step -
accuracy: 0.8163 - loss: 0.4113
Epoch 400/400
11/11             0s 2ms/step -
accuracy: 0.8141 - loss: 0.4111
```

[39]:
```python
# Visualisation
plt.plot(history.history["loss"])
plt.title("Training loss")
plt.xlabel("Epoch")
plt.ylabel("Binary crossentropy")
plt.show()

plt.plot(history.history["accuracy"])
plt.title("Training accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.show()
```

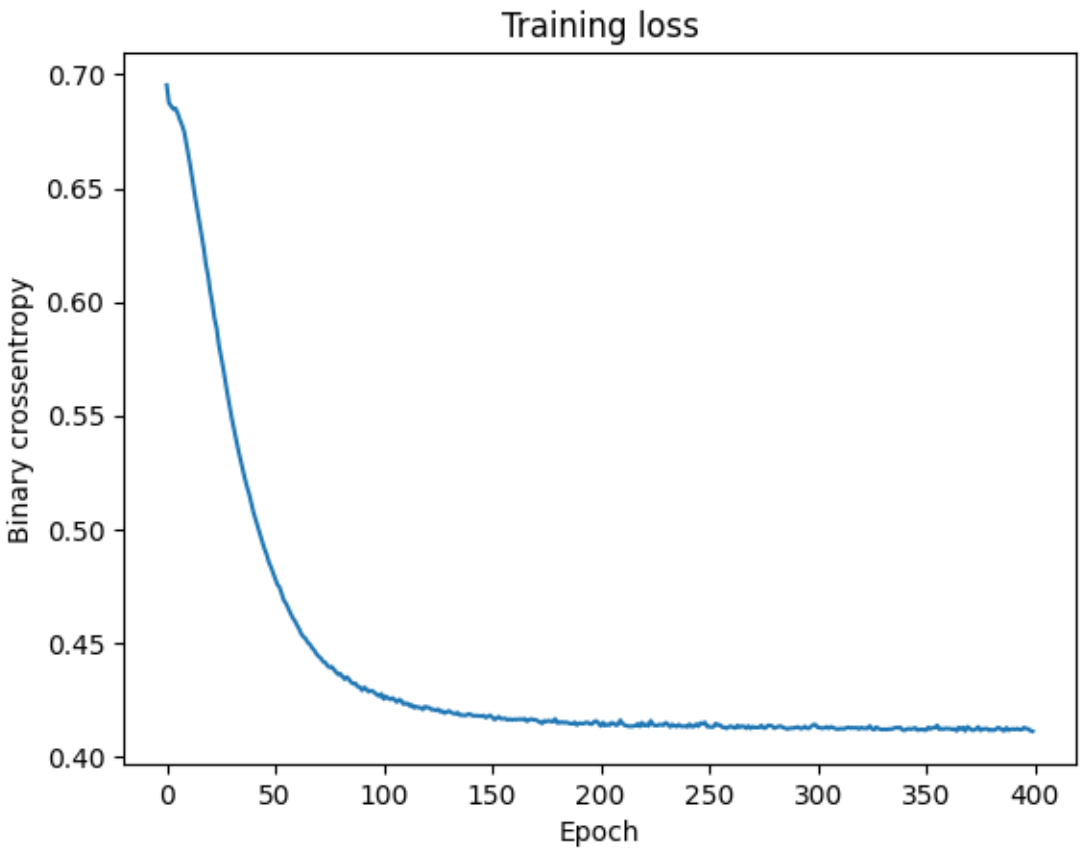
```
<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

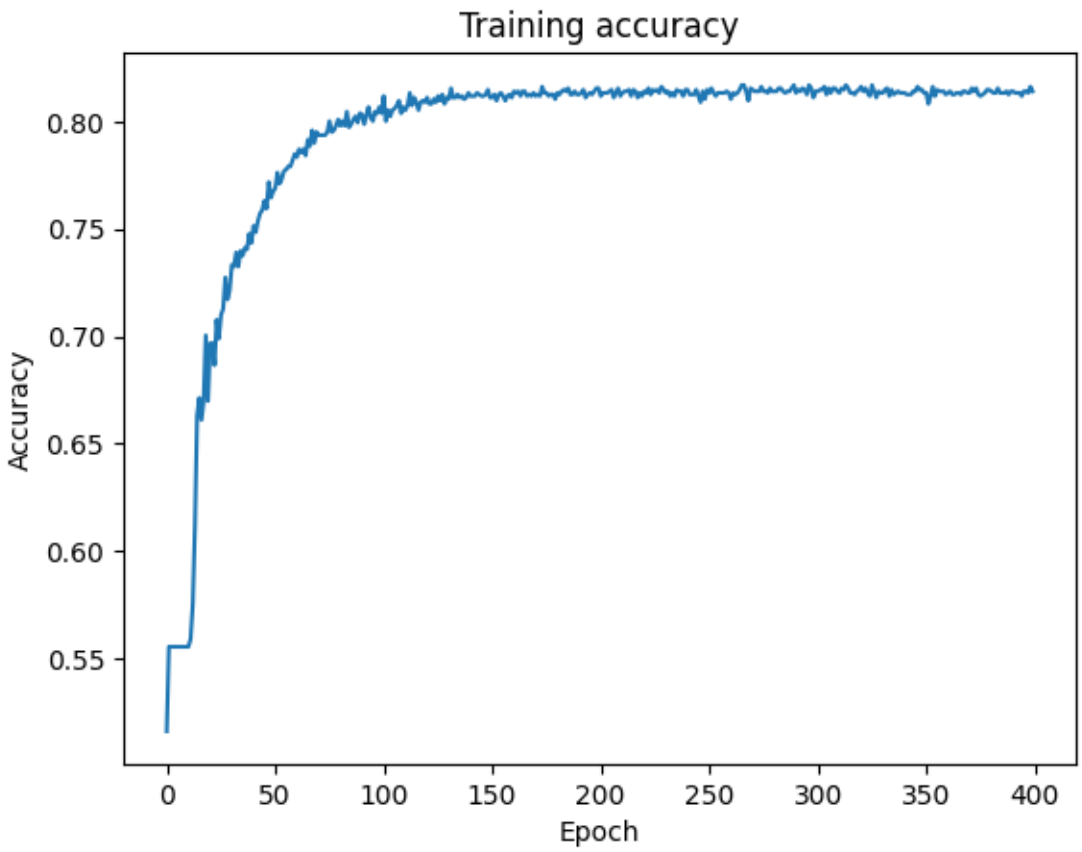
<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>
```




```
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
```



We achieved better result, this time the accuracy reached 0.81.

1.5 6.5

[45]:
```
model.predict(np.reshape(np.array([0.0,0.1]),(1,2)))
```

```
1/1             0s 20ms/step
```

[45]:
```
array([[0.29521906, 0.7047809 ]], dtype=float32)
```

[50]:
```
x1grid, x2grid = np.meshgrid(x1list, x2list)

grid_points = np.c_[x1grid.ravel(), x2grid.ravel()]
Z = model.predict(grid_points)[:, 0]
Z = Z.reshape(x1grid.shape)

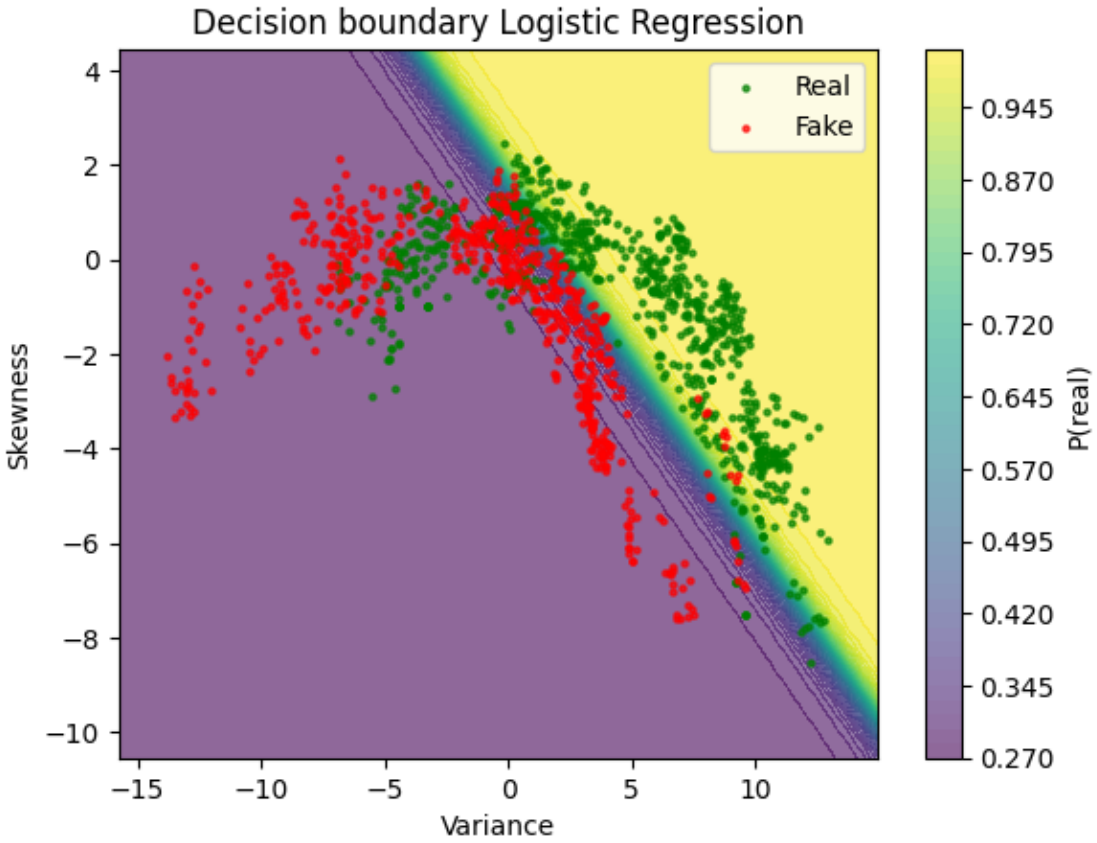
plt.contourf(x1grid, x2grid, Z, levels=50, cmap="viridis", alpha=0.6)
plt.colorbar(label="P(real)")

plt.scatter(real[:, 0], real[:, 1], color="green", s=5, alpha=0.7, label="Real")
plt.scatter(fake[:, 0], fake[:, 1], color="red",   s=5, alpha=0.7, label="Fake")
```

```
plt.title("Decision boundary Logistic Regression")
plt.xlabel("Variance")
plt.ylabel("Skewness")
plt.legend()
plt.show()
```

```
79/79             0s 438us/step
```

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1.6 Comparison: Single Neuron vs Multi-Layer Model

The single neuron model achieved ~70% accuracy, limited by its linear decision boundary. The multi-layer model improves significantly by learning a non-linear boundary, visible in the contour plot, with higher confidence predictions for both classes.

Some misclassifications remain mostly real banknotes predicted as fake (false negatives).